

September 25 Quiz

I understand that I must take the quizzes only face-to-face in my assigned Discussion section using Honorlock and my webcam. The exam will be online, so I will need to bring my charged device with me to take the face-to-face exam. Scratch paper will be provided. No books or notes or other outside material can be used during the exam.

- ☒ A. True
- ☐ B. False





Ask Your Neighbor

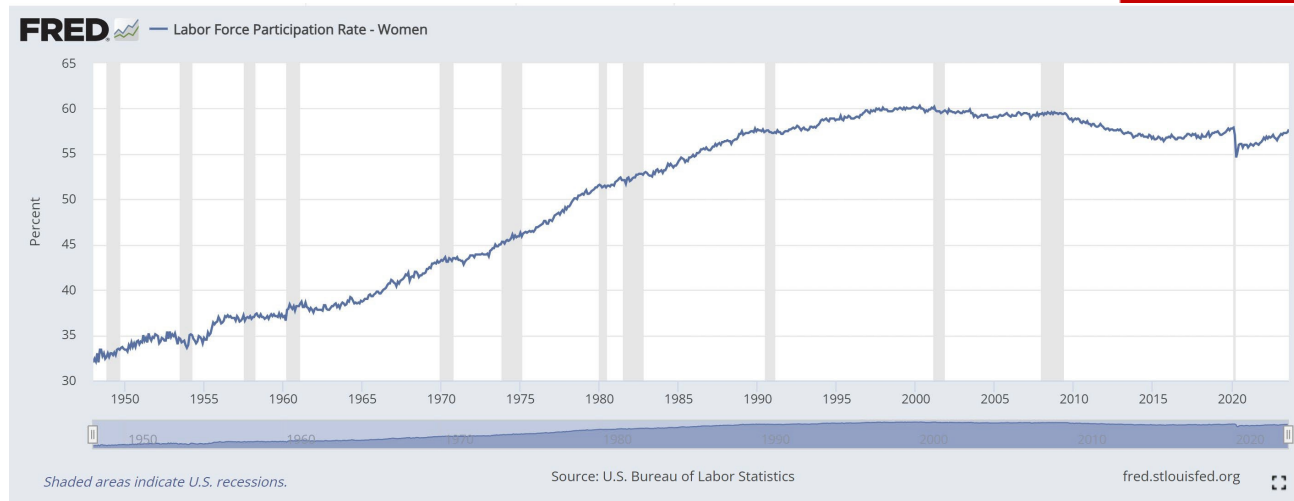
The rise in women's labor force participation rates 1965-1995 might be explained by

- a. Greater materialism
- b. Improved job opportunities for women
- c. Efficiency gains in household work that freed up more time for market work
- d. Any or all of the above

$U = a \ln c + b \ln l$ } MRS
a ↑ } flatter

w(1-t) ↑ sub effects
income effects

like h ↑





Ask your neighbor

If an increase in after-tax real wages causes leisure to decrease, then we know that

- A. Substitution effects dominated Income effects
- B. Substitution effects were dominated by Income effects
- C. Substitution and Income effects were equal and opposite in sign
- D. None of the above—there isn't enough information to know

Get this result with quasi-linear U

$$U = aC + b \ln l$$

↑ ↑

C and l enter differently
(not both log) so $w(c-t)$

doesn't cancel out and
 l is a function $w(c-t)$

Pure income effect $l \uparrow$ $C \uparrow$
 $\underbrace{\hspace{1cm}}$
 normal goods
 if $l \uparrow$ $N \downarrow$

Ask Your Neighbor

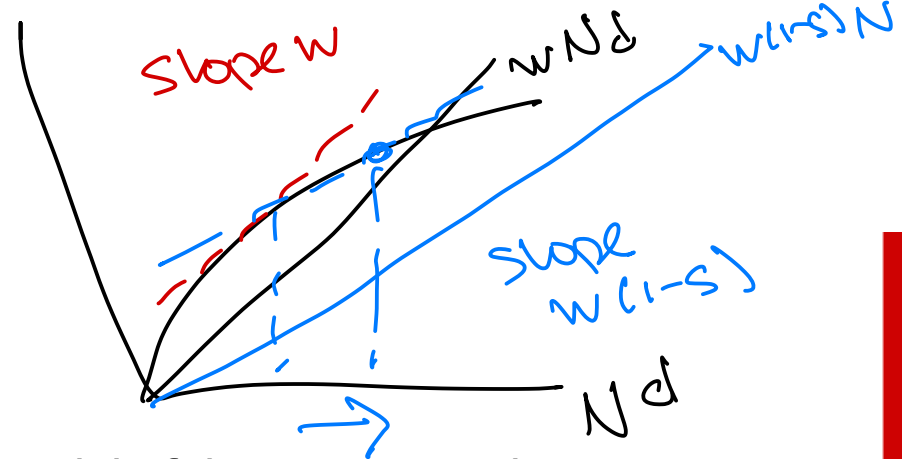
According to the Chapter 4 One-Period model of the representative household, an increase in asset values increasing “profits” would most likely lead to a _____ in labor input and a _____ in Consumption.

$\pi \uparrow$
 $(T \downarrow)$

- a. Decrease; decrease
- b. Decrease; increase**
- c. Increase; decrease
- d. Increase increase



$$w(1-s) < w$$



Ask Your Neighbor

According to the Chapter 4 one-period model of the representative firm, a proportional government wage subsidy will lead to

- a. A decrease in both labor demand and production
- b. An increase in both labor demand and production
- c. No change in either labor demand and production
- d. An ambiguous change in both labor demand and production

as $N \uparrow$
 $\sim \uparrow$

$$w(1-s) < w = MPN_{\text{original at } N}$$

$$w(1-s) < MPN$$

choose N_d so

that $MPN \downarrow$ to equal
 $w(1-s) \Rightarrow N_d \uparrow$ $MPN \downarrow$
 (diminishing returns)



- slope $IC = MRS = \frac{MU(L)}{MU(C)} \uparrow$ } choose more leisure

Steeper IC

tangent to the right of initial IC

Ask your neighbor

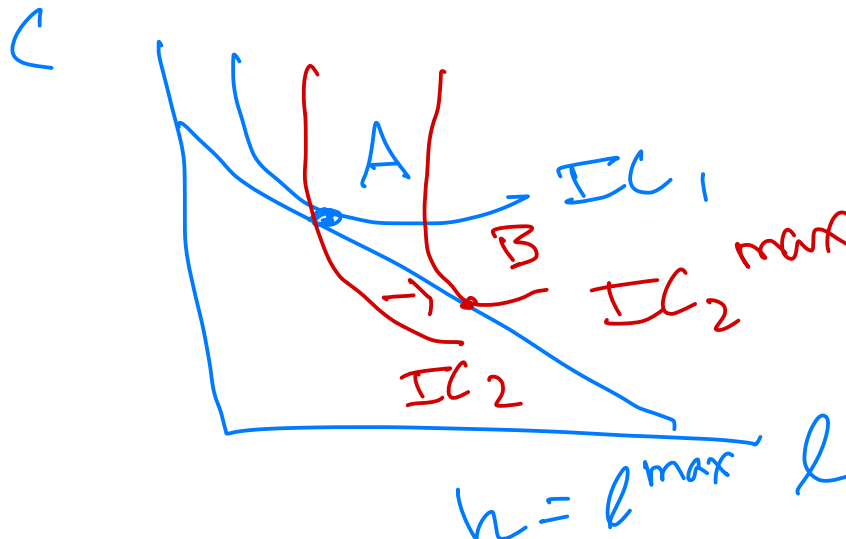
Consider the following theoretical utility function:

$$U = a \ln C + b \ln(l)$$

Assume an increase in the coefficient b on leisure. All other things being equal, this will most likely IC

- A. Increase the forecasted estimate for Consumption C
- B. Increase the forecasted estimate for labor supply N
- C. Increase the forecasted estimate for leisure l
- D. Have no impact as income and substitution effects will offset.

$\Delta U ?$



$$\pi - \tau = \phi$$





Ask your neighbor

Consider the following one-period theoretical utility function:

Quasi-linear utility

$$U(C, l) = aC + b \ln(l)$$

as $w(l-t) \uparrow$ $l \downarrow$
 $N \uparrow$

substitution effects dominate income effects

All other things being equal, this model will most likely predict _____ labor input at low levels of Income and _____ labor input at high levels of wage income

- ~~A. Lower; lower~~
- ☒ B. Lower; higher
- C. higher; lower
- ~~D. Higher; higher~~



Ask your neighbor

Consider the following one-period theoretical utility function:

$$U = C + b \ln(l)$$

} Quasi-linear utility

where

$$C = w(1 - t)(h - l)$$

All other things being equal, an increase in after-tax real wages will cause leisure to

- ☒ A. Fall (substitution effect dominates the income effect)
- ☐ B. rise (income effect dominates the substitution effect)
- ☐ C. be unchanged (income and substitution effects cancel)
- ☐ D. Anything is possible

$$MRS = \frac{MU(l)}{MU(c)} = \frac{b/l}{1} = \frac{b}{l}$$

$$\frac{b}{w(1-t)} = l \quad \left\{ \begin{array}{l} \frac{b}{l} = MRS = w(1-t) \\ \text{as } w(1-t) \uparrow \quad l \downarrow \end{array} \right.$$



Ask Your Neighbor

Based on the theory and data from class, if economists use a log-linear utility function (below) to forecast the business-cycle responses of labor supply to a **decline** in the real wage, they will likely _____ both employment and GDP

$$U = a \ln C + b \ln(I)$$

- ☒ A. overestimate
- ☐ B. underestimate
- ☐ C. perfectly estimate
- ☐ D. be unable to estimate

income and substitution
effects cancel so
assumes $\bar{N}$ as
 $\Delta[w(t-t^*)]$

over business cycle { In data correlation $(w(t-t^*), N) > 0$
if $w(t-t^*) \downarrow \Rightarrow N \downarrow$



Ask Your Neighbor

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$$U = a \ln C + b \ln(I)$$

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